

# Siam Arefin

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## PROFILE

Junior AI/ML Engineer with hands-on experience in Machine Learning, LLMs, Generative AI, Computer Vision, and MLOps. Skilled in Python, FastAPI, and Docker, with a strong software engineering foundation.

## EDUCATION

### BSc in Software Engineering

Shahjalal University of Science and Technology  
03/2022 – 07/2026 | Sylhet, Bangladesh

## SKILLS

### Problem Solving

- Solved 800+ problems on Codeforces  and other competitive programming platforms

### Programming

- Python, C, C++, Java, JavaScript

### AI/ML

- Machine Learning, Deep Learning, NLP, CNNs, RNNs, Transformers, Scikit-learn, PyTorch, TensorFlow, Feature Engineering

### LLM & Generative AI

- LLMs, Prompt Engineering, Fine-tuning, RAG, Vector Databases, LangChain, AI Agents, Hugging Face, Seq2Seq, Tokenization, Diffusion Models, Google Gemini API

### Computer Vision

- OpenCV, YOLO, Object Detection, Image Processing, Image Generation, Sim2Real, NMS, Ensemble Learning

### Backend & Full-Stack

- FastAPI, REST APIs, Node.js, Express.js, React.js, MongoDB, JWT, Authentication, SQL

### Software Engineering & MLOps

- OOP, SOLID, Design Patterns, DSA, Software Architecture, Unit Testing, Git/GitHub, Docker, CI/CD, Azure

## COMPETITIONS

- Multi-Instance Object Detection Challenge  - 1st place International competition hosted by a USA startup
- ML Olympiad - CO2 Emissions Prediction Challenge  - 1st runner-up
- DL Enigma 1.0  - SUST CSE Carnival 2024 - 4th place
- DL Sprint 3.0 | Bengali AI Math Olympiad  - Top 10

## EXPERIENCE

### Software Engineer (ML team) Intern

Kite Games Studio, Mohakhali DOHS, Dhaka (Onsite)

11/2025 – 01/2026 | Bangladesh

- Conducted R&D in image processing, machine learning, and generative AI for production-oriented applications.
- Worked with Diffusion and FLUX-based generative AI models for image generation and processing.
- Key contributor to an Android application with 5M+ downloads, currently reaching 700K+ daily active users.

## RESEARCH EXPERIENCE

### omicML: An Integrative Tool of Bioinformatics and Machine Learning Algorithms to Identify Transcriptomic Biomarkers

12/2024 – 10/2025 | Bangladesh

Collaboration with Cross-Functional Teams – Department of Biochemistry and Molecular Biology (BMB), SUST

Role: Machine Learning Research Engineer

- Implemented scalable ML deployment and monitoring pipelines using MLOps best practices.
- Collaborated with bioinformatics researchers to integrate computational modeling with biological large data analysis.

## PROJECTS

### HealthMonitor – AI-Powered Health Monitoring App

- Developed and deployed a MERN-stack web app with custom-trained ML model for CVD prediction
- Integrated the Google Gemini API for intelligent advice
- Enabled secure login and signup with JWT-based authentication
- Developed a RAG-based system to deliver personalized insights using medical records and health-related data

### Multi-Instance Object Detection

- Merged synthetic datasets from multiple sources and applied ensemble learning to reduce the Sim2Real domain gap
- Implemented custom postprocessing (NMS, confidence filtering) and fine tuned YOLO model to maximize generalization to real-world images

### AI Processing API — AI/ML Backend System

- Designed a modular AI backend with layered system architecture, Provider/Adapter pattern, and Dependency Injection.
- Built REST APIs for audio transcription and medical document OCR, with validation, structured extraction, testing, and Dockerized deployment.